

S&S QUIZ-9 SOLUTION

Sol(1): Given signal,

$$x(t) = e^{-5t} u(t-t_0) + e^{-\beta t} u(t) \quad \text{--- (1)}$$

Part(1): As we know that — $x(t) \xrightarrow[\text{IFT}]{\text{FT}} X(s)$

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-st} dt \\ &= \int_{-\infty}^{\infty} e^{-5t} u(t-t_0) e^{-st} dt + \int_{-\infty}^{\infty} e^{-\beta t} u(t) e^{-st} dt \\ &= \int_{t_0}^{\infty} e^{-(s+5)t} dt + \int_0^{\infty} e^{-(s+\beta)t} dt \\ &= \left. \frac{e^{-(s+5)t}}{-(s+5)} \right|_{t_0}^{\infty} + \left. \frac{e^{-(s+\beta)t}}{-(s+\beta)} \right|_0^{\infty} \\ &= \frac{e^{-\infty} - e^{-(s+5)t_0}}{-(s+5)} + \frac{e^{-\infty} - e^{-0}}{-(s+\beta)} \\ &= \frac{e^{-(s+5)t_0}}{s+5} + \frac{1}{s+\beta} \quad \rightarrow (3 \text{ Marks}) \end{aligned}$$

$$\text{ROC: } \sigma > -5$$

$$\text{ROC: } \sigma > -\beta$$

$\rightarrow (2 \times 0.5 \text{ Marks})$

For common ROC,

Case(I): $\beta < 5$

Common ROC, $\sigma > -\beta$

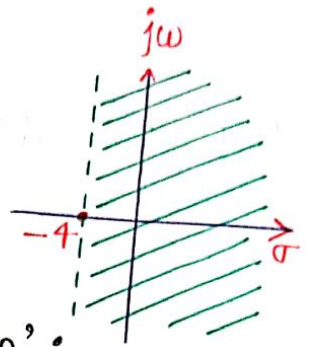
$\rightarrow (1.5 \text{ Marks})$

Case(II) $\beta \geq 5$

Common ROC, $\sigma > -5$

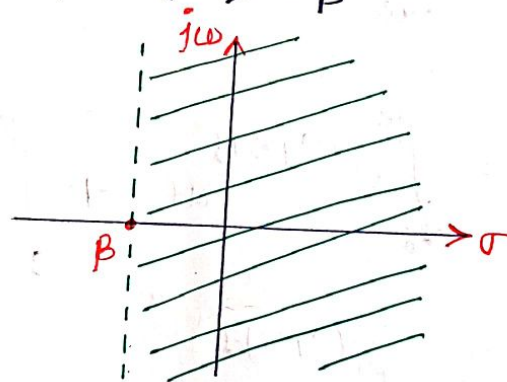
→ (1.5 Marks)

Part(2) $\div$ Given ROC of $X(s)$, $\text{Real}\{s\} > -4$
 $\sigma > -4$



constraints on the real & imaginary part of ' β ' :

Case(I) $\div$ Common ROC, $\sigma > -\beta$



Here, $\text{Real}\{\beta\} = 4$

→ (2 Marks)

$\text{Im}\{\beta\} = \text{No constraints}$

→ (1.0 Marks)